

Mathematics Behind the Options Calculator

Pheneas Newman

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Throughout, T is time remaining in years. Calendar-time theta is $\Theta = -\partial V/\partial T$, with S , K , r , q , and σ held fixed. Theta is reported per year; vega and rho are derivatives per unit change in decimal volatility and interest rate (divide by 100 for a percentage point).

1 European Call

$$C = Se^{-qT} \Phi(d_1) - Ke^{-rT} \Phi(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$$\boxed{Se^{-qT} \phi(d_1) = Ke^{-rT} \phi(d_2)} \quad (\text{Black-Scholes identity})$$

C Option price

S Spot price

K Strike price

q Dividend rate

T Time to maturity

r Risk-free rate

σ Volatility

Φ Standard normal CDF

ϕ Standard normal pdf

1.1 Delta

$$\Delta = \frac{\partial C}{\partial S} = e^{-qT} \Phi(d_1)$$

$$\begin{aligned} \Delta &= \frac{\partial C}{\partial S} = e^{-qT} \Phi(d_1) + S e^{-qT} \phi(d_1) \left(\frac{\partial d_1}{\partial S} \right) - K e^{-rT} \phi(d_2) \left(\frac{\partial d_2}{\partial S} \right) \\ &= e^{-qT} \Phi(d_1) + \underbrace{S e^{-qT} \phi(d_1) \frac{1}{S \sigma \sqrt{T}} - K e^{-rT} \phi(d_2) \frac{1}{S \sigma \sqrt{T}}}_{=0} \\ \Delta &= e^{-qT} \Phi(d_1) \end{aligned}$$

1.2 Gamma

$$\Gamma = \frac{\partial^2 C}{\partial S^2} = e^{-qT} \left(\phi(d_1) \frac{1}{S \sigma \sqrt{T}} \right)$$

$$\begin{aligned} \Gamma &= \frac{\partial^2 C}{\partial S^2} = \frac{\partial \Delta}{\partial S} \\ &= e^{-qT} \frac{\partial}{\partial S} (\Phi(d_1)) \\ \Gamma &= e^{-qT} \phi(d_1) \frac{1}{S \sigma \sqrt{T}} \end{aligned}$$

1.3 Vega

$$\mathcal{V} = \frac{\partial C}{\partial \sigma} = S e^{-qT} \phi(d_1) \sqrt{T}$$

$$\begin{aligned} \mathcal{V} &= \frac{\partial C}{\partial \sigma} = S e^{-qT} \phi(d_1) \left(\frac{\partial d_1}{\partial \sigma} \right) - K e^{-rT} \phi(d_2) \left(\frac{\partial d_2}{\partial \sigma} \right) \\ \frac{\partial d_1}{\partial \sigma} &= \frac{\partial}{\partial \sigma} \left(\frac{\ln(\frac{S}{K})}{\sigma \sqrt{T}} + \frac{T(r-q)}{\sigma \sqrt{T}} \right) + \frac{1}{2} \sqrt{T} \\ \frac{\partial d_2}{\partial \sigma} &= \frac{\partial}{\partial \sigma} \left(\frac{\ln(\frac{S}{K})}{\sigma \sqrt{T}} + \frac{T(r-q)}{\sigma \sqrt{T}} \right) - \frac{1}{2} \sqrt{T} \\ \frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma} &= \sqrt{T} \\ \mathcal{V} &= S e^{-qT} \phi(d_1) \sqrt{T} \end{aligned}$$

1.4 Theta

$$\Theta = -\frac{\partial C}{\partial T} = -e^{-qT} \frac{S \phi(d_1) \sigma}{2 \sqrt{T}} - r K e^{-rT} \Phi(d_2) + q S e^{-qT} \Phi(d_1)$$

$$\begin{aligned}
\frac{\partial C}{\partial T} &= S \left(-qe^{-qT} \Phi(d_1) + e^{-qT} \phi(d_1) \frac{\partial d_1}{\partial T} \right) - K \left(-re^{-rT} \Phi(d_2) + e^{-rT} \phi(d_2) \frac{\partial d_2}{\partial T} \right) \\
\frac{\partial d_1}{\partial T} &= \frac{(\sigma^2 + 2r - 2q)T - 2 \ln(\frac{S}{K})}{4\sigma T^{\frac{3}{2}}} := A \\
\frac{\partial d_2}{\partial T} &= \frac{(\sigma^2 + 2r - 2q)T - 2 \ln(\frac{S}{K})}{4\sigma T^{\frac{3}{2}}} - \frac{\sigma}{2\sqrt{T}} := A - \frac{\sigma}{2\sqrt{T}} \\
&= Se^{-qT}(-q\Phi(d_1) + \phi(d_1)A) - Ke^{-rT}(-r\Phi(d_2) + \phi(d_2)(A - \frac{\sigma}{2\sqrt{T}})) \\
&= -qSe^{-qT}\Phi(d_1) + rKe^{-rT}\Phi(d_2) + Ke^{-rT}\phi(d_2)\frac{\sigma}{2\sqrt{T}} + \underbrace{A(Se^{-qT}\phi(d_1) - Ke^{-rT}\phi(d_2))}_{=0} \\
&= Ke^{-rT}\phi(d_2)\frac{\sigma}{2\sqrt{T}} + rKe^{-rT}\Phi(d_2) - qSe^{-qT}\Phi(d_1)
\end{aligned}$$

Using the Black-Scholes identity $Se^{-qT}\phi(d_1) = Ke^{-rT}\phi(d_2)$, then applying $\Theta = -\partial C/\partial T$:

$$\Theta = -e^{-qT} \frac{S\phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}\Phi(d_2) + qSe^{-qT}\Phi(d_1)$$

1.5 Rho

$$\boxed{\rho = \frac{\partial C}{\partial r} = KTe^{-rT}\Phi(d_2)}$$

$$\begin{aligned}
\rho &= \frac{\partial C}{\partial r} = Se^{-qT}(\phi(d_1)\frac{\partial d_1}{\partial r}) - K(-Te^{-rT}\Phi(d_2) + e^{-rT}\phi(d_2)\frac{\partial d_2}{\partial r}) \\
\frac{\partial d_1}{\partial r} &= \frac{T}{\sigma\sqrt{T}} = \frac{\sqrt{T}}{\sigma} = \frac{\partial d_2}{\partial r} \\
&= Se^{-qT}\phi(d_1)\frac{\sqrt{T}}{\sigma} + KTe^{-rT}\Phi(d_2) - Ke^{-rT}\phi(d_2)\frac{\sqrt{T}}{\sigma} \\
&= \frac{\sqrt{T}}{\sigma} \left(\underbrace{Se^{-qT}\phi(d_1) - Ke^{-rT}\phi(d_2)}_{=0} \right) + KTe^{-rT}\Phi(d_2) \\
\rho &= KTe^{-rT}\Phi(d_2)
\end{aligned}$$

3 Binary Call (Cash-or-Nothing)

$$C_{\text{bin}} = Q e^{-rT} \Phi(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

C_{bin} Option price

Q Fixed cash payout if $S_T > K$

S Spot price

K Strike price

q Dividend rate

T Time to maturity

r Risk-free rate

σ Volatility

Φ Standard normal CDF

ϕ Standard normal pdf

3.1 Delta

$$\Delta = \frac{\partial C_{\text{bin}}}{\partial S} = Q e^{-rT} \frac{\phi(d_2)}{S\sigma\sqrt{T}}$$

$$\begin{aligned} \Delta &= \frac{\partial C_{\text{bin}}}{\partial S} = Q e^{-rT} \phi(d_2) \frac{\partial d_2}{\partial S} \\ &= Q e^{-rT} \phi(d_2) \frac{1}{S\sigma\sqrt{T}} \end{aligned}$$

3.2 Gamma

$$\Gamma = \frac{\partial^2 C_{\text{bin}}}{\partial S^2} = -Q e^{-rT} \frac{d_1 \phi(d_2)}{S^2 \sigma^2 T}$$

$$\begin{aligned}
\Gamma &= \frac{\partial^2 C}{\partial S^2} = Qe^{-rT} \left(-\phi(d_2) \frac{1}{S^2 \sigma \sqrt{T}} + \frac{1}{S \sigma \sqrt{T}} \cdot \frac{\partial \phi(d_2)}{\partial S} \right) \\
\frac{\partial \phi(d_2)}{\partial S} &= \frac{\partial \phi(d_2)}{\partial d_2} \cdot \frac{\partial d_2}{\partial S} = -d_2 \phi(d_2) \frac{1}{S \sigma \sqrt{T}} \\
&= Qe^{-rT} \left(\frac{1}{S \sigma \sqrt{T}} \left(-d_2 \phi(d_2) \frac{1}{S \sigma \sqrt{T}} \right) - \phi(d_2) \frac{1}{S^2 \sigma \sqrt{T}} \right) \\
&= -Qe^{-rT} \frac{1}{S \sigma \sqrt{T}} \left(d_2 \phi(d_2) \frac{1}{S \sigma \sqrt{T}} + \phi(d_2) \frac{1}{S} \right) \\
&= -Qe^{-rT} \frac{\phi(d_2)}{S \sigma \sqrt{T}} \left(\frac{d_2}{S \sigma \sqrt{T}} + \frac{\sigma \sqrt{T}}{S \sigma \sqrt{T}} \right) \\
&= -Qe^{-rT} \frac{\phi(d_2)}{S^2 \sigma^2 T} (d_2 + \sigma \sqrt{T}) \\
\Gamma &= -Qe^{-rT} \frac{d_1 \phi(d_2)}{S^2 \sigma^2 T}
\end{aligned}$$

3.3 Vega

$$\boxed{\mathcal{V} = \frac{\partial C_{\text{bin}}}{\partial \sigma} = -Qe^{-rT} \frac{d_1}{\sigma} \phi(d_2)}$$

$$\begin{aligned}
\mathcal{V} &= \frac{\partial C}{\partial \sigma} = Qe^{-rT} \phi(d_2) \cdot \frac{\partial d_2}{\partial \sigma} \\
&= Qe^{-rT} \phi(d_2) \left(\frac{\partial d_1}{\partial \sigma} - \sqrt{T} \right) \\
\frac{\partial d_1}{\partial \sigma} &= -\frac{\ln\left(\frac{S}{K}\right) + T(r - q)}{\sigma^2 \sqrt{T}} + \frac{1}{2} \sqrt{T} \\
&= Qe^{-rT} \phi(d_2) \left(-\frac{\ln\left(\frac{S}{K}\right) + T(r - q)}{\sigma^2 \sqrt{T}} - \frac{1}{2} \sqrt{T} \right) \\
&= -Qe^{-rT} \phi(d_2) \left(\frac{\ln\left(\frac{S}{K}\right) + T(r - q)}{\sigma^2 \sqrt{T}} + \frac{1}{2} \sqrt{T} \right) \\
&= -Qe^{-rT} \phi(d_2) \underbrace{\left(\frac{\ln\left(\frac{S}{K}\right) + T(r - q) + \frac{1}{2} \sigma^2 T}{\sigma^2 \sqrt{T}} \right)}_{= \frac{d_1}{\sigma}} \\
&= -Qe^{-rT} \phi(d_2) \frac{d_1}{\sigma}
\end{aligned}$$

4 Binary Put (Cash-or-Nothing)

$$\begin{aligned}P_{\text{bin}} &= Q e^{-rT} \Phi(-d_2) \\d_1 &= \frac{\ln\left(\frac{S}{K}\right) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \\d_2 &= d_1 - \sigma\sqrt{T}\end{aligned}$$

P_{bin} Option price

Q Fixed cash payout if $S_T < K$

S Spot price

K Strike price

q Dividend rate

T Time to maturity

r Risk-free rate

σ Volatility

Φ Standard normal CDF

ϕ Standard normal pdf

4.1 Delta

$$\Delta = \frac{\partial P_{\text{bin}}}{\partial S} = -Q e^{-rT} \frac{\phi(d_2)}{S \sigma \sqrt{T}}$$

Same derivation as call but negative

4.2 Gamma

$$\Gamma = \frac{\partial^2 P_{\text{bin}}}{\partial S^2} = Q e^{-rT} \frac{d_1 \phi(d_2)}{S^2 \sigma^2 T}$$

Same derivation as call

4.3 Vega

$$\mathcal{V} = \frac{\partial P_{\text{bin}}}{\partial \sigma} = Q e^{-rT} \frac{d_1}{\sigma} \phi(d_2)$$

Same derivation as call

